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In Sight

A Window To Change

Here's a look at what the next version of Microsoft Windows offers

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When the successor to the Windows XP throne was being touted, seemingly eons ago, it promised heaven on earth—a pass to the utopian kingdom of computing, wherein everyone is happy and malware happens only to other people.

As with any promise, reality eroded the ideal, and product delays forced Microsoft to freeze upon a more realistic goal for their next-generation operating system. This product was then code-named Longhorn, and now answers to the marketing moniker of Windows Vista.

Although not the product it once was, Vista still offers a vast landscape of features. Much has happened since Windows XP was released—built on the Windows NT kernel, XP was never prepared for the Internet at large, nor for the several mini-revolutions which have since shook the technology space.

Vista hopes to provide a fertile ground and a secure ecosystem within which seemingly disparate elements will thrive—the WWW, wired P2P networks, wireless networks, ad-hoc networks, high-capacity hard drives, high-density displays, cell phones, mobile audio and video players, the rise of the laptop, the proliferation of the LCD screen, and our ever-increasing need to communicate.

Here, we offer a bird's-eye view of the immediate changes Vista will bring to our computing experience, along with a superficial look at some of the underlying technologies driving that change.

Defining The New Look

The most visible—and thus the most talked about—feature of Vista is its new look. Both with Vista and its Xbox 360 gaming console, Microsoft is betting on the High-Definition (HD) era to come. HD will bring in displays with a much higher dots-per-inch density (dpi) than is currently offered by typical monitors. The idea is to offer fonts, icons and other user interface elements that will scale according to the size and density offered by your monitor. So, if you're one of the lucky few to possess a widescreen 24-inch LCD monitor, Vista will offer

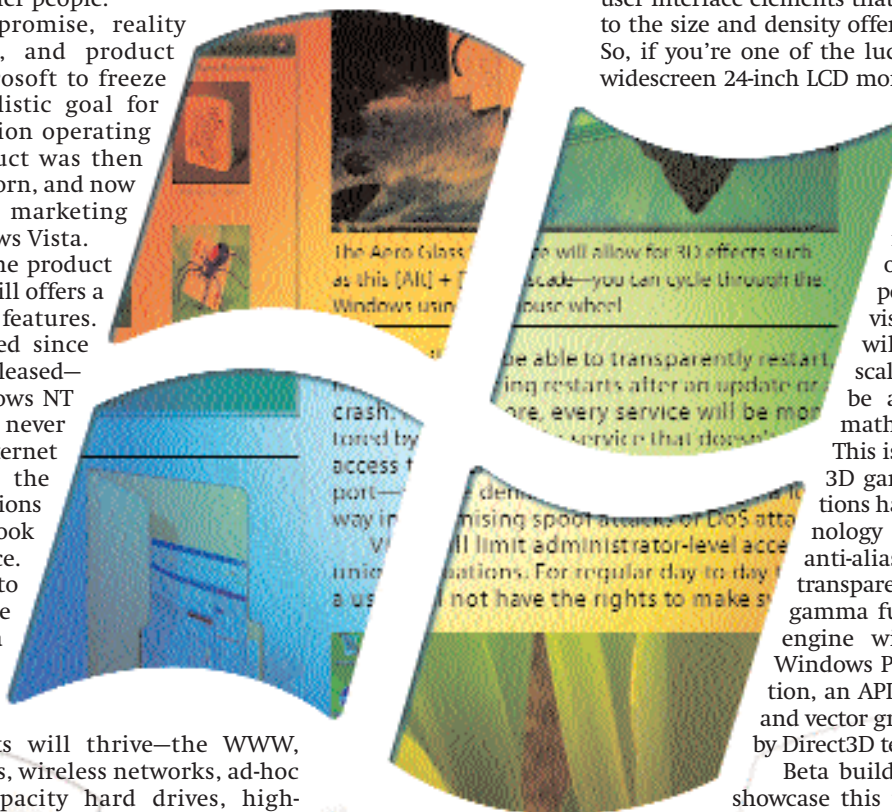
larger, more pleasant fonts and icons.

This is done by using a vector-based graphics engine, replacing the bitmap or raster engine that powers XP. Since every visual element on Vista will be a vector image, scaling and resizing will be a simple matter of maths, hardware willing. This is very similar to how 3D games or Flash animations handle data. This technology will bring in text anti-aliasing, animations, transparencies, and advanced gamma functions. The vector engine will be part of the Windows Presentation Foundation, an API built on XML, .NET, and vector graphics, and powered by Direct3D technology.

Beta builds of Vista currently showcase this technology through the Aero Glass interface, which offers transparent window borders (hence “glass”), drop shadows, scalable icons and fonts, and animations.

Information On The Side

Microsoft has embraced XML, and a modified variant of that data markup language called XAML will be ubiquitous under Vista. The Windows Sidebar, which will perhaps share



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